

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

02

10 answers · Rationale-rich · Teach from doc

Q1**231**

The correct answer is 231. It's given that $\sqrt{k-x} = 58-x$. Squaring both sides of this equation yields $k-x = (58-x)^2$, which is equivalent to the given equation if $58-x > 0$. It follows that if a solution to the equation $k-x = (58-x)^2$ satisfies $58-x > 0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k-x = (58-x)^2$ can be rewritten as $k-x = 3,364 - 116x + x^2$. Adding x to both sides of this equation yields $k = x^2 - 115x + 3,364$. Subtracting k from both sides of this equation yields $0 = x^2 - 115x + (3,364 - k)$. The number of solutions to a quadratic equation in the form $0 = ax^2 + bx + c$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. Substituting -115 for b , 1 for a , and $3,364 - k$ for c in $b^2 - 4ac$ yields $(-115)^2 - 4(1)(3,364 - k)$, or $4k - 231$. The equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution if the discriminant is equal to zero, or $4k - 231 = 0$. Subtracting 231 from both sides of this equation yields $4k = 231$. Dividing both sides of this equation by 4 yields $k = 57.75$. Therefore, if $k = 57.75$, then the equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution. Substituting 57.75 for k in this equation yields $0 = x^2 - 115x + (3,364 - 57.75)$, or $0 = x^2 - 115x + 3,306.25$, which is equivalent to $0 = (x - 57.5)^2$. Taking the square root of both sides of this equation yields $0 = x - 57.5$. Adding 57.5 to both sides of this equation yields $57.5 = x$. To check whether this solution satisfies $58 - x > 0$, the solution, 57.5 , can be substituted for x in $58 - x > 0$, which yields $58 - 57.5 > 0$, or $0.5 > 0$. Since 0.5 is greater than 0 , it follows that if $k = 57.75$, or $4k = 231$, then the given equation has exactly one real solution. If $4k < 231$, then the discriminant, $4k - 231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is 231 .

Q2

C

C. There is exactly 1 solution.

Choice C is correct. The second equation of the system can be rewritten as $y = 5x - 8$.

Substituting $5x - 8$ for y in the first equation gives $5x - 8 = x^2 + 3x - 7$. This equation can be solved as shown below:

$$x^2 + 3x - 7 - 5x + 8 = 0$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0$$

$$x = 1$$

Substituting 1 for x in the equation $y = 5x - 8$ gives $y = -3$. Therefore, $(1, -3)$ is the only solution to the system of equations.

Choice A is incorrect. In the xy -plane, a parabola and a line can intersect at no more than two points. Since the graph of the first equation is a parabola and the graph of the second equation is a line, the system cannot have more than 2 solutions. Choice B is incorrect. There is a single ordered pair (x, y) that satisfies both equations of the system. Choice D is incorrect because the ordered pair $(1, -3)$ satisfies both equations of the system.

Q3**17.5**

Accept also: 35/2

The correct answer is $\frac{35}{2}$. Subtracting the second equation from the first equation yields $y + k - y - k = x + 26 - x^2 - 5x$, or $2k = -x^2 + 6x + 26$. This is equivalent to $x^2 - 6x + 2k - 26 = 0$. It's given that the system has exactly one distinct real solution; therefore, this equation has exactly one distinct real solution. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one distinct real solution when the discriminant, $b^2 - 4ac$, is equal to 0. The equation $x^2 - 6x + (2k - 26) = 0$ is of this form, where $a = 1$, $b = -6$, and $c = 2k - 26$. Substituting these values into the discriminant, $b^2 - 4ac$, yields $-6^2 - 4(1)(2k - 26)$. Setting the discriminant equal to 0 yields $-6^2 - 4(1)(2k - 26) = 0$, or $-36 - 8k + 104 = 0$, or $-8k + 68 = 0$. Subtracting 68 from both sides of this equation yields $-8k = -68$. Dividing both sides of this equation by -8 yields $k = \frac{35}{2}$. Note that 35/2 and 17.5 are examples of ways to enter a correct answer.

Q4**D****D. Zero**

Choice D is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $5x^2 + 10x + 16 = 0$, $a = 5$, $b = 10$, and $c = 16$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $10^2 - 4(5)(16)$, or $100 - 320$, or -220 . Since the value of its discriminant is negative, the given equation has zero real solutions. Therefore, the number of distinct real solutions the given equation has is zero.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5**A****A.** 6

Choice A is correct. It's given that the graph of $y = x^2 - 9$ and line p intersect at $(1, a)$ and $(5, b)$. Therefore, the value of y when $x = 1$ is the value of a, and the value of y when $x = 5$ is the value of b. Substituting 1 for x in the given equation yields $y = (1)^2 - 9$, or $y = -8$. Similarly, substituting 5 for x in the given equation yields $y = (5)^2 - 9$, or $y = 16$. Therefore, the intersection points are $(1, -8)$ and $(5, 16)$. The slope of line p is the ratio of the change in y to the change in x between these two points: $\frac{16 - (-8)}{5 - 1} = \frac{24}{4}$, or 6.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors in determining the values of a, b, or the slope of line p.

Q6**D****D.** $\frac{26}{5}$

Choice D is correct. By the definition of absolute value, if $-5x + 13 = 73$, then $-5x + 13 = 73$ or $-5x + 13 = -73$. Subtracting 13 from both sides of the equation $-5x + 13 = 73$ yields $-5x = 60$. Dividing both sides of this equation by -5 yields $x = -12$. Subtracting 13 from both sides of the equation $-5x + 13 = -73$ yields $-5x = -86$. Dividing both sides of this equation by -5 yields $x = \frac{86}{5}$. Therefore, the solutions to the given equation are -12 and $\frac{86}{5}$, and it follows that the sum of the solutions to the given equation is $-12 + \frac{86}{5}$, or $\frac{26}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is a solution, not the sum of the solutions, to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**9**

The correct answer is 9. The given equation can be rewritten as $54 - x = 25$. Dividing each side of this equation by 5 yields $4 - x = 5$. By the definition of absolute value, if $4 - x = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting 4 from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting 4 from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and 9, the positive solution to the given equation is 9.

Q8**-1**

The correct answer is -1. A quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when its discriminant, $b^2 - 4ac$, is equal to 0. In the given equation, $-16x^2 - 8x + c = 0$, $a = -16$ and $b = -8$. Substituting -16 for a and -8 for b in $b^2 - 4ac$ yields $-8^2 - 4(-16)c$, or $64 + 64c$. Since the given equation has exactly one solution, $64 + 64c = 0$. Subtracting 64 from both sides of this equation yields $64c = -64$. Dividing both sides of this equation by 64 yields $c = -1$. Therefore, the value of c is -1.

Q9**A****A. $(1 + \sqrt{3}, \sqrt{3})$**

Choice A is correct. The solution to the given system of equations can be found by solving the first equation for x , which gives $x = y + 1$, and substituting that value of x into the second equation which gives $y + 1 + y = (y + 1)^2 - 3$. Rewriting this equation by adding like terms and expanding $(y + 1)^2$ gives $2y + 1 = y^2 + 2y - 2$. Subtracting $2y$ from both sides of this equation gives $1 = y^2 - 2$. Adding 2 to both sides of this equation gives $3 = y^2$. Therefore, it follows that $y = \pm\sqrt{3}$. Substituting $\sqrt{3}$ for y in the first equation yields $x - \sqrt{3} = 1$. Adding $\sqrt{3}$ to both sides of this equation yields $x = 1 + \sqrt{3}$. Therefore, the ordered pair $(1 + \sqrt{3}, \sqrt{3})$ is a solution to the given system of equations.

Choice B is incorrect. Substituting $\sqrt{3}$ for x and $-\sqrt{3}$ for y in the first equation yields $\sqrt{3} - (-\sqrt{3}) = 1$, or $2\sqrt{3} = 1$, which isn't a true statement. Choice C is incorrect. Substituting $1 + \sqrt{5}$ for x and $\sqrt{5}$ for y in the second equation yields $(1 + \sqrt{5}) + \sqrt{5} = (1 + \sqrt{5})^2 - 3$, or $1 + 2\sqrt{5} = 2\sqrt{5} + 3$, which isn't a true statement. Choice D is incorrect. Substituting $\sqrt{5}$ for x and $(-1 + \sqrt{5})$ for y in the second equation yields $\sqrt{5} + (-1 + \sqrt{5}) = (\sqrt{5})^2 - 3$, or $2\sqrt{5} - 1 = 2$, which isn't a true statement.

Q10

D

D. Zero

Choice D is correct. A point x, y is a solution to a system of equations if it lies on the graphs of both equations in the xy -plane. In other words, a solution to a system of equations is a point x, y at which the graphs intersect. It's given that the first equation is $y = 18$. Substituting 18 for y in the second equation yields $18 = -3x - 18^2 + 15$. Subtracting 15 from each side of this equation yields $3 = -3x - 18^2$. Dividing each side of this equation by -3 yields $-1 = x - 18^2$. Since the square of a real number is at least 0, this equation can't have any real solutions. Therefore, the graphs of the equations intersect at zero points.

Alternate approach: The graph of the second equation is a parabola that opens downward and has a vertex at 18, 15. Therefore, the maximum value of this parabola occurs when $y = 15$. The graph of the first equation is a horizontal line at 18 on the y -axis, or $y = 18$. Since 18 is greater than 15, or the horizontal line is above the vertex of the parabola, the graphs of these equations intersect at zero points.

Choice A is incorrect. The graph of $y = 15$, not $y = 18$, and the graph of the second equation intersect at exactly one point.

Choice B is incorrect. The graph of any horizontal line such that the value of y is less than 15, not greater than 15, and the graph of the second equation intersect at exactly two points.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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